NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Rare Is Not Never

AP Statistics · Unit 2 · Topic 2.10 · B: 2026-11-05 · E: 2026-11-25

[STAR] TODAY'S PROBLEM

Suppose a basketball player has made 70 percent of her free throws over a long career. Tonight she makes 3 of 10 attempts. Let X be the number she would make in 10 attempts if her chance on each shot were still 0.70. The graph shows the binomial model as expected frequencies per 1,000 sets of 10 shots.

Claim A: "She is in a slump—making only 3 of 10 never happens at her career rate."

Claim B: "With only 10 shots, 3 makes is ordinary chance variation."

FIRST TAKE — before the video (5 min)

Which claim, if either, do you trust more? Cite one detail from the result or graph.

[BOOK] BINS BEFORE BINOMIAL (keep on the page)

A binomial random variable counts successes across trials that are **B**inary, plausibly **I**ndependent, fixed in **N**umber, and have the **S**ame success probability. Define X, n, and p before calculating. $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$. A cumulative event such as $X \leq 3$ includes every value from 0 through 3. A small probability makes an outcome unusual under the model; it does not make it impossible or name a cause.

Finish after the video:

DOK 1 **(a)** Define the random variable X in context, state n and p, and write a probability expression for exactly 3 makes.

DOK 2 **(b)** Calculate $P(X = 3)$ using the binomial probability formula and interpret the result in context.

DOK 3 **(c)** Check all four binomial conditions for this model, explicitly assessing whether independence is plausible. Then compute $P(X \leq 3)$ and adjudicate Claims A and B: which claim, if either, is warranted, and why? State what additional shooting evidence would change your conclusion.

Frame: The binomial model is __________ because __________________________, although independence requires ____________________.

Frame: Since $P(X \leq 3) =$ ________, Claim ____ is _________, but ________________________.

EXIT — turn this in

One thing the video changed about my first take: __________________________________